

Benchmarking 3DVar with Different Optimization Methods

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OSSE Configuration

Model and DA Method

Model:

- Lorenz 96
- Isca from GFDL (TBD)
T21 aquaplanet: 64*32*20
(with only surface observation)

DA Method:

- 3D-Var

NMC...

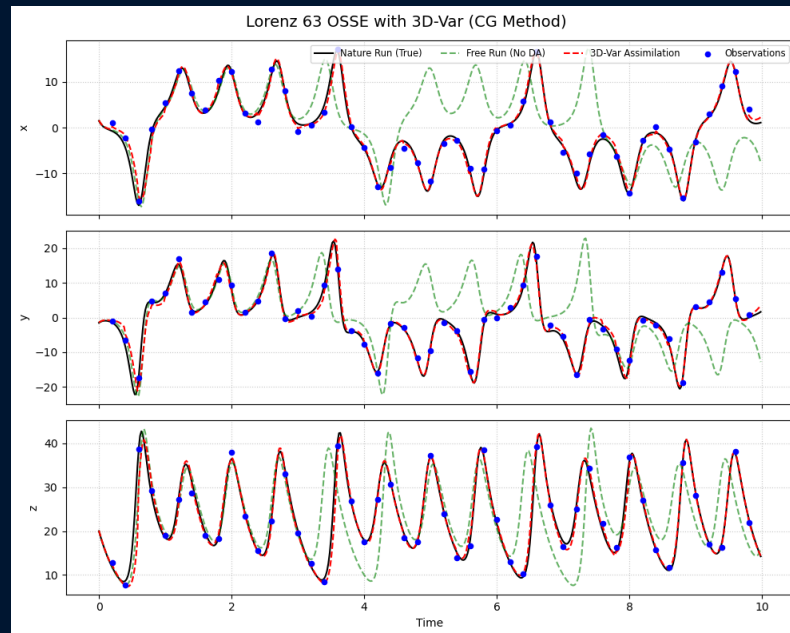
FGAT...

Using three optimization methods

Conjugate Gradient Method

Optimization Method 1

- I will write a CG algorithm by my self :D



OSSE based on Lorenz 63 system,
3D-Var with conjugate gradient method.

Sailfish Optimization (SFO)

Optimization Method 2

- Inspired by the collective hunting behavior of sailfish
- Sardines as potential solutions
- Chasing between sailfish and sardines can avoid the local minima problem



Harris Hawks Optimization (HHO)

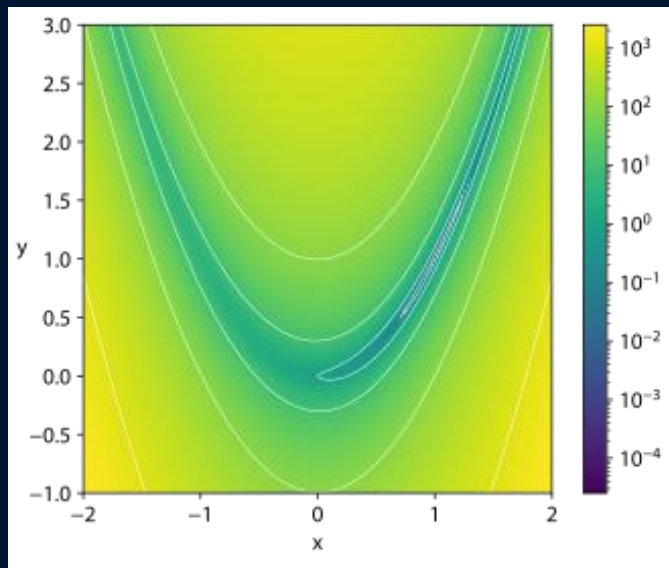
Optimization Method 3

- Inspired by the hunting behavior of Harris's Hawk and escape mechanism of rabbit
- Rabbit as predicted solution
- Chasing between hawks and rabbit can avoid the local minima problem

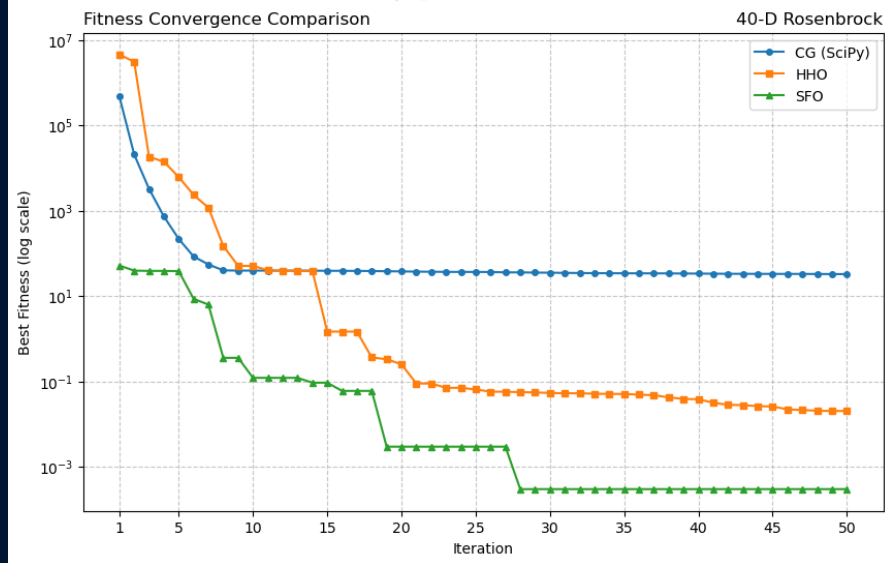


Test

40-D Rosenbrock function



$$f(\mathbf{x}) = f(x_1, x_2, \dots, x_N) = \sum_{i=1}^{N/2} [100(x_{2i-1}^2 - x_{2i})^2 + (x_{2i-1} - 1)^2]$$



Objectives

- Evaluate the performance of 3D-Var with different optimization algorithms: efficiency and accuracy
- Whether the FGAT / NMC method can reduce accuracy differences among algorithms

Reference

1. S. Shadravan et al. *"The Sailfish Optimizer: A novel nature-inspired metaheuristic algorithm for solving constrained engineering optimization problems,"* Engineering Applications of Artificial Intelligence, Volume 80, 2019, Pages 20-34, ISSN 0952-1976.
2. Ali Asghar Heidari et al. *"Harris hawks optimization: Algorithm and applications,"* Future Generation Computer Systems, Volume 97, 2019, Pages 849-872, ISSN 0167-739X.

THANKS!